

Dynamic programming

WHEN?

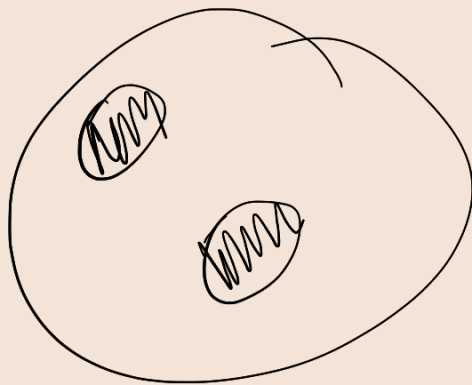
- the solution is a result of a sequence of decisions
- principle of optimality

ABOUT:

- polynomial run-time
- optimal solution
- $\neq$ Greedy
- combines solutions from sub-problems
- optimisation to recursion
- memorises subproblem solutions

Principles:

- ① optimality
- ② overlapping subproblems $\sim$
- ③ memoization $\sim$ cache



Optimality:

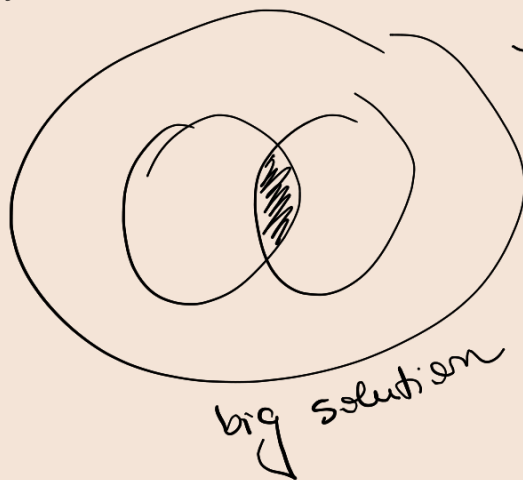
↳ the solution to the main problem's subproblems

are themselves optimal solutions to their own subproblems.

! The value of the optimal sol. is recursively defined.



Overlapping Subproblems — the problem can be broken down into subprob. that are overlapping,



for each we calculate the optimal solution



assemble

Memorization

↳ store the solutions to the sub-problems for later.

1, 1, 2, 3, 5, 8, 13, ... Fibonacci